

Medicinal Plant Identification Using CNN and Pytorch Lightning

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ABSTRACT - Medicinal plants have long been integral to healthcare, providing natural therapeutic compounds. With the growing interest in alternative medicine, accurate identification of these plants is crucial. Traditional methods of plant identification rely on expert botanists and manual examination, which are both time-consuming and susceptible to errors. This paper seeks to develop a reliable system for identifying medicinal plants through the use of Convolutional Neural Networks (CNNs) and the PyTorch Lightning framework. Our approach leverages deep learning to automate plant identification, addressing the increasing demand for herbal medicine authentication. Using a dataset of annotated medicinal plant images, we design a CNN architecture to extract features and accurately classify species. Transfer learning fine-tunes pre-trained models to improve classification performance, even when training data is limited. Performance indicators including precision, recall, and F1-score are used to assess the system. The goal is to create a reliable tool for botanists, pharmacists, and herbal medicine practitioners, aiding in the authentication and preservation of traditional medicinal knowledge. This automated identification system promises to enhance the precision and effectiveness of the classification of medicinal plants, supporting sustainable use and conservation efforts.

Keywords: Deep Learning, Image Processing, CNN

I. INTRODUCTION

Medicinal plants represent a vital component of traditional medicine systems worldwide, offering a diverse array of bioactive compounds with therapeutic properties. From ancient herbal remedies to modern pharmaceuticals, these plants have played a crucial role in treating various ailments and improving human health [3]. However, the sustainable management of medicinal plant resources is increasingly threatened by habitat loss, overexploitation, and climate change. Moreover, the manual identification of plant species is often challenging and subjective, requiring specialized knowledge and expertise. Advances in deep learning, particularly with regard to computer vision, have revolutionized image analysis and identification tasks over the past years [5][6][7]. In particular, the Convolutional Neural Networks have shown remarkable ability to automatically identify objects and patterns in images. Researchers have achieved unprecedented levels of accuracy and efficiency in tasks like object recognition, classification or segmentation through the development of these networks on a massive SVM, KNN, and Naïve Bayes, are proposed for accurate plant identification based on distinct features [8].

amount of annotated image data. Inspired by these advancements, our project seeks to apply deep learning techniques to the domain of medicinal plant detection [10] [11] [13]. Our objective is to develop an autonomous system that helps to identify and classify medicinal plant species from botanical images with high accuracy and efficiency. Our approach seeks to overcome the limitations of traditional techniques and deliver a flexible solution for conservationists, researchers as well as healthcare professionals through using CNNs trained in large datasets on plant images. The key components of our methodology include data collection, preprocessing, model training, and evaluation. A diverse dataset of botanical images featuring medicinal plant species from various regions has been curated. By iterative training and optimization, we aim to enhance the model's capacity to recognize distinctive features of these plants.

II. RELATED WORK

The identification of herbal plants using computer vision and deep learning techniques has become more and more common in recent years. Comparing with the conventional methods of plant identification, e.g. manually inspecting plants and analysing their morphology, these methods have been shown to be faster and more precise.

One of the approaches to identify the medicinal plants is via Convolutional Neural Networks (CNNs). CNN algorithm helps in classifying the images of the plants based on the different traits of the leaf, including its form, colour and so on [12].

The study combines CNNs and Log-Gabor filters in plant identification, achieving 98% accuracy with DenseNet201 and OTAMNet on medicinal plant data. Transfer learning is utilized, showcasing the efficacy of pretrained networks [1].

The study presents traits of the leaf, including its form, that integrates EfficientNetB1 with mobile app interfaces. While achieving notable accuracy enhancements, the dynamic database integrates both public and private plant datasets[2]. Deep learning models, including InceptionV3, MobileNet, and Xception, are evaluated for their accuracy in recognizing medicinal plants from leaf images. MobileNet emerges as the optimal choice, demonstrating superior performance in identifying 11 medicinal plants based on leaf features [4]. Computer vision neural network techniques, including CNN,

Computer vision neural network techniques, including CNN, SVM, KNN, and Naïve Bayes, are proposed for accurate plant identification based on distinct features. Following validation,

the project achieved promising results with a precision of 92.3% and an F1 score of 90.56%, highlighting the effectiveness of the methods used. [9].

The proposed approach combines a Conditional Generative Adversarial Network, Convolutional Neural Network, and Logistic Regression, achieving an average accuracy of 96.1% across seven datasets. Results, ranging from 99.0% to 100% on specific datasets, showcase the effectiveness of combining DL and logistic regression in plant leaf recognition, surpassing alternative techniques [20].

In order to quickly and accurately identify the medicinal plant *Simplicia* from microscopic fragments, the “MikrobatX” method, using artificial intelligence, combines Convolutional Neural Networks CNN with SIFT feature extraction. Achieving a notable 89.42% accuracy on the Mikrobat dataset, it proves effective in overcoming identification challenges, suggesting potential for further optimization [21].

Based on the above references, it's evident that there's a significant focus on making use of deep learning methods, specifically convolutional neural networks, to identify and classification of medicinal plants. Several studies have examined the integration of deep learning into additional technologies such as image processing, transfer learning and wireless applications to increase accuracy and applicability in practice. These advancements hold promise for various applications, including traditional medicine, biodiversity conservation, agricultural sustainability, and drug discovery. Additionally, the challenges such as dataset scarcity, model optimization, and real-time performance are addressed through innovative methodologies and optimization techniques. Overall, the potential of deep learning to change medicinal plant identification and its diverse applications across various domains is highlighted by research in this area.

III. METHODOLOGY

One kind of deep neural network is a convolutional neural network (CNN), which is primarily used for the analysis of visual images. CNNs are intended for automatic and adaptive learning of the spatial hierarchies of features from input images.

There are six different steps involved in the methodology like collecting and preparing datasets, the model architecture and so on.

A DATA COLLECTION AND PREPARATION:

We took a diverse dataset of images containing different species of medicinal plants.

We ensured that the dataset is labelled accurately with the corresponding plant species.

Then performed data cleaning to remove any corrupted or irrelevant images.

B DATA PREPROCESSING:

Then, for input to the CNN model, each image was shrunk in a uniform size that is usually round.

We've been using some data enhancement techniques like rotation, flip and spin. Cropping techniques to increase variability in data sets and improve generalization of models.

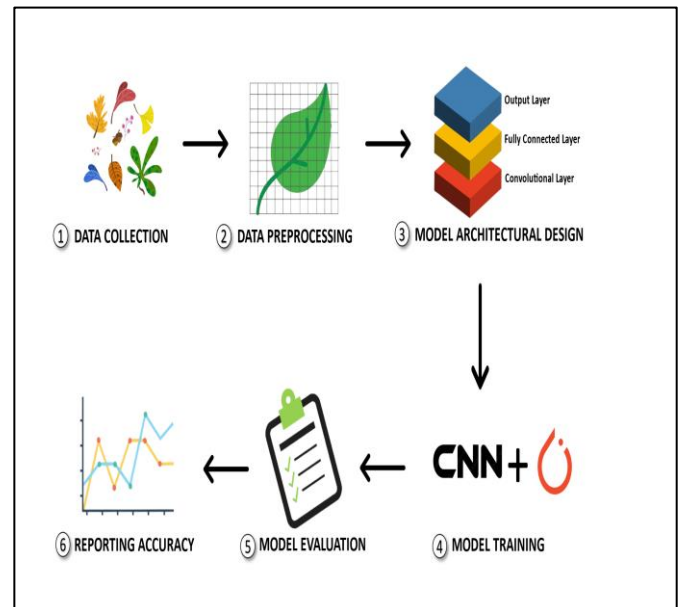


Fig. 1. Architecture diagram for plant identification

C MODEL ARCHITECTURE DESIGN:

Two convolutional layers, two fully linked layers, and one output layer make up the model.

Input: RGB images with dimensions of 3, batch size, height, width.

Convolutional layer 1: Apply 6 filters of size 3x3 to an input image, then activate ReLU and maxpool.

Convolutional layer 2: Use 16 filters of size 3x3 in the output of the first convolutional layer, followed by ReLU activation and max pooling. Three completely interconnected layers with activation functions ReLU layer are available.

The output layer is the final fully connected layer that corresponds to the number of outputs units corresponding to the number of classes, followed by a SoftMax activation for obtaining class probabilities.

D MODEL TRAINING:

Implement the designed CNN architecture using PyTorch Lightning, providing a modular and efficient training framework.

Define appropriate loss functions, typically categorical cross-entropy for multi-class classification tasks.

To facilitate effective training, configure optimizer parameters such as learning speed, acceleration and weight reduction.

To avoid overwriting, train the model using a training dataset and monitor performance of validation set.

To optimise convergence of training and avoid overfill, use techniques like learning rate timing and starting at the earliest possible time.

E MODEL EVALUATION AND OPTIMIZATION:

The performance of the trained model is evaluated based on the testing of the dataset to assess its generalization capabilities.

To determine the performance of classification, make advantage of evaluation metrics like F1 score, recall, accuracy, and precision. Visualize confusion matrices to gain insights into model behaviour and identify potential misclassifications.

Conduct additional analysis such as class-wise performance evaluation to identify strengths and weaknesses of the model across different plant species.
Regularize the model by applying techniques such as dropout or L2 regularization to mitigate overfitting.

F REPORTING ACCURACY:

Report the final accuracy achieved by the trained model on the test dataset, providing a quantitative measure of its classification performance.

Include confidence intervals or standard deviations to indicate the variability in performance estimates.

To contextualize the model's performance in the taxonomy of medicinal plants, compare the achieved accuracy with existing methods or benchmarks.

We can develop a robust CNN model for the classification of medicinal plants, achieve high accuracy and demonstrate its efficacy in real-world applications by carefully following this methodology.

IV. DATASET

We have collected commonly used leaves of plants and herbs with medicinal properties from this database. In the dataset for training purposes, which included more than 3000 images and used a few types of medicinal plants to test, we had about 40 different kinds of medicinal plant species.

To investigate the model, Figure 2 displays some of the more than 2097 photos took of 14 different plant species for testing.



Fig. 2. Sample pictures in the dataset

V. IMPLEMENTATION

Convolutional neural networks, which are the CNN algorithm and the Pytorch lightning, are the algorithm we used here to identify medicinal plants. Figure 3 provides a clear indication of the steps involved in implementing it, as well as an explanation of how they work:

Importing Libraries: The necessary libraries and modules include PyTorch, PyTorch Lightning, Matplotlib, and some others.

2. Define Transformations: We used transformation techniques to make the images suitable for the system to clearly analyse the image with techniques like horizontal flip, rotation, resizing and cropping the image.

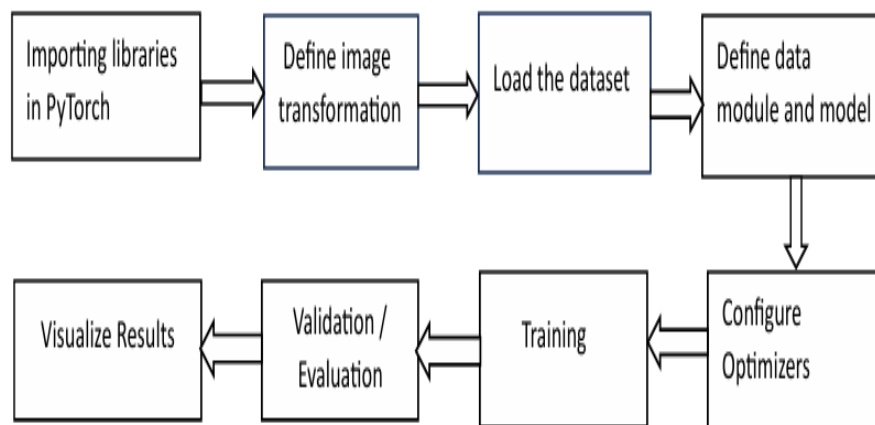


Fig. 3. Steps involved in the implementation process

3. Load Dataset: we loaded our plant dataset using PyTorch's 'ImageFolder' dataset class in which we placed the images we collected.

4. Define Data Module: we defined a PyTorch Lightning 'DataModule' class to organize the data loading and preprocessing steps. This class should include methods for setting up the data and creating data loaders such as the self.Transform, self.batch_size and self.train methods.

5. Defining Model: we defined the neural network model as a PyTorch Lightning 'Lightning Module' class. This class contains the model architecture ('_init_' method) and forward pass implementation ('forward' method).

6. Configure Optimizer: The optimizer 'Adam' and its associated hyper parameters is configured in the 'configure_optimizers' method of the Lightning Module.

7. Define Training and Validation Steps: The 'training_step' and 'validation_step' methods in the LightningModule class contains training a batch, accuracy calculation, loss calculation, and logging.

8. Defining Testing Step: This step is optional and we used this step to evaluate our model based on a separate test set by implementing the 'test_step' method in the LightningModule class.

9. Training the Model: we created an instance for the model and a trainer object for maximum of 20 epochs and then, called

the ‘fit’ method of the trainer object, passing in our model and data module.

10. *Evaluate the Model*: We used the trainer object's test method to evaluate the model's performance in a set of tests after training.

11. *Visualize Results*: We can visualize the results, such as training loss, validation accuracy, and test metrics, using Matplotlib or other visualization library.

12. *Make Predictions*: if we want to make predictions on new data, we can load the trained model and pass the new data through the model's forward pass method.

VI. RESULTS AND ANALYSIS

After the process of executing codes in the jupyter notebook with appropriate plant datasets, modules and methods, the epochs the model runs at is 19 and the performance metrics are also calculated and displayed after the end of execution and they are:

Precision: precision is the percentage of the samples predicted as a certain plant is being correctly classified.

Recall: It indicates that the model is capable of properly identifying samples which are positively identified.

f1-Score: The f1 score offers a single metric that balances precision and recall by taking the mean of the two. An increase in the F1 score indicates better model performance.

Support: This is the number of actual occurrences for each class in the test database. For example, in the test database there are 146 samples of "Aloe vera".

Using PyTorch, the model was 90.33% accurate as a whole. Figure 4 shows the performance indicators for each of the plants in the data set, with their respective bar charts as shown in Figure 5.

TABLE 1: ACCURACY PERCENTAGES, PRECISION, RECALL, F1 SCORE, AND SUPPORT DATA SETS.

Plant	Precision	Recall	f1-Score	Support
Aloe Vera	0.7308	0.7037	0.7170	27
Amla	0.9062	0.9062	0.9062	32
Arali	0.8810	1.0000	0.9367	37
Ashwagandha	0.9630	1.0000	0.9811	26
Betel	0.8667	0.8387	0.8525	31
Guava	1.0000	0.7618	0.8649	21
Henna	0.6970	0.7667	0.7302	30
Hibiscus	0.7698	0.9091	0.8333	33
Lemon	0.9545	0.9545	0.9545	22
Mint	0.8636	0.8636	0.8636	22
Neem	0.9600	0.9231	0.9412	26
Papaya	1.0000	0.6944	0.8197	36
Pepper	0.7941	0.9310	0.8571	29
Tulasi	0.6857	0.8261	0.7451	23
Accuracy			0.9033	0.8148
Macro avg	0.9271	0.8683	0.8683	29
Weighted avg	0.7570	0.6930	0.7510	29

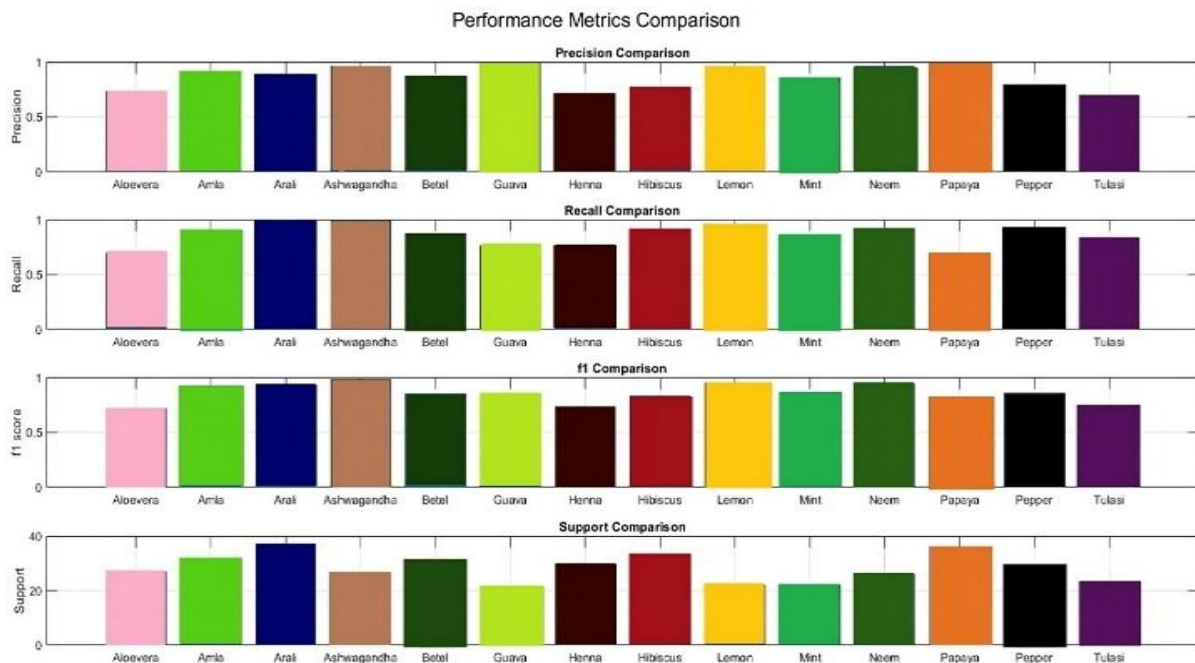


Fig. 4. A bar chart showing the performance metrics of all 14 plant species.

VII. CONCLUSIONS AND FUT URE WORK

Convolutional neural networks (CNNs) have shown remarkable potential in the field of medicinal plants through

their use of image datasets containing plant images, enabling researchers, pharmacists and botanists to accurately classify and identify medical products. In this area, the ability of CNNs

to identify relevant features from images by means of automatic extraction makes them a powerful and reliable tool that can be used regardless of differences in lighting, background or image quality. Despite the promising advancements made in medicinal plant identification using CNNs, there are a number of avenues for research and improvement, such as expanding data sets, fine tuning architecture, advanced data enhancement, integration with mobile applications or better quality assurance.

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